

Billing & Folio Module v1

1. Purpose

The Billing / Folio module provides the financial ledger backbone for a guest stay. It records all monetary charges and adjustments related to a reservation in an immutable, auditable manner. Payments, invoices, and taxes are explicitly out of scope for v1.

2. Scope (v1)

Included: - Folio creation (manual, ADMIN only) - Line items (service charges, minibar, discounts, adjustments) - Strict tenant isolation - RBAC enforcement - Immutable ledger design - Folio lifecycle (OPEN → CLOSED)

Excluded: - Payments - Refunds - Taxes / GST - Invoices - Auto room-night charge generation

3. Core Entities

3.1 Folio

Represents a single financial account for a reservation.

Key characteristics: - One folio per reservation - Tenant-scoped - Cannot be deleted - Cannot be reopened after closure

Lifecycle: - OPEN - CLOSED (terminal)

3.2 Folio Line Item

Represents an immutable financial event.

Types: - ROOM_CHARGE (future use) - SERVICE_CHARGE - MINIBAR - DISCOUNT - ADJUSTMENT

Rules: - Line items are append-only - No updates or deletes allowed - Totals are derived (quantity × unit price)

4. RBAC Matrix

Folio

Action	ADMIN	STAFF
Create folio	✓	✗
View folio	✓	✓
Close folio	✓	✗

Line Items

Action	ADMIN	STAFF
Add service/minibar	✓	✓
Add discount/adjustment	✓	✗
Modify/delete	✗	✗

5. Business Rules

- A folio must belong to exactly one tenant
- A folio must reference a valid reservation
- Line items cannot be added once folio is CLOSED
- Only ADMIN can close a folio
- Discounts and adjustments require ADMIN role
- Currency is fixed to INR (v1 default)

6. API Endpoints

Create Folio (ADMIN)

POST /folios

Request: - reservationId (UUID)

Get Folio by Reservation

GET /folios/:reservationId

Roles: - ADMIN - STAFF

Add Line Item

POST /folios/:folioId/line-items

Roles: - SERVICE_CHARGE, MINIBAR → ADMIN, STAFF - DISCOUNT, ADJUSTMENT → ADMIN only

Close Folio

PATCH /folios/:folioId/close

Role: - ADMIN only

7. Error Scenarios

- 404: Folio or reservation not found
 - 403: RBAC violation
 - 400: Invalid state transition (adding to closed folio)
 - 409: Duplicate folio for reservation
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8. Runtime Test Cases (curl + jq)

8.1 Create Folio (ADMIN)

```
curl -s -X POST http://localhost:3000/folios
-H "Authorization: Bearer $ADMIN_TOKEN"
-H "Content-Type: application/json"
-d '{ "reservationId": "<UUID>" }' | jq
```

Expected: - 201 - Folio object

8.2 Add Service Charge (STAFF)

```
curl -s -X POST http://localhost:3000/folios/<FOLIO_ID>/line-items
-H "Authorization: Bearer $STAFF_TOKEN"
-H "Content-Type: application/json"
-d '{
  "type": "SERVICE_CHARGE",
  "description": "Airport Pickup",
  "quantity": 1,
  "unitPrice": 75,
  "currency": "INR"
}' | jq
```

Expected: - totalAmount = 75

8.3 Add Discount (STAFF – FAIL)

Expected: - 403 Forbidden

8.4 Add Discount (ADMIN – PASS)

Expected: - totalAmount negative - postedBy.role = ADMIN

8.5 Close Folio (ADMIN)

Expected: - status = CLOSED - closedAt populated

8.6 Add Line Item After Close (FAIL)

Expected: - 400 Bad Request - Message: Cannot add items to closed folio

9. Lock Criteria

The module is considered LOCKED when: - All above test cases pass at runtime - No cross-tenant access is possible - RBAC is enforced for all write operations - Folio immutability is preserved

10. Status

Billing / Folio Module v1 — LOCKED & PRODUCTION READY